

Project - Northwind Sales & Operations Audit

Project overview -: "Our sales grew in 2014, but our management team is flying blind. We are seeing high revenue, but we don't know who our most loyal customers are, which regions are struggling, or if our shipping partners are actually delivering on time. Furthermore, we have a lot of 'missing data' in our shipping logs, and we are worried this reflects a breakdown in our warehouse operations."

Data Structure

- Rows: Thousands of transaction records (Order IDs ranging from 1001 to 1432+).
- Columns: 20+ fields including:
- Order details (Order ID, Order Date, Shipped Date)
- Customer info (Customer ID, Name, Address, City, State, Country)
- Salesperson & Region
- Shipping details (Shipper Name, Ship Name, Ship Address, Ship City)
- Payment type
- Product details (Name, Category, Unit Price, Quantity, Sales)

Key Insights

- Regions Covered: West, East, North, South (USAbased).
- Payment Types: Cash, Check, Credit Card.
- Top Categories:
- Beverages (Coffee, Tea, Beer, Chai)
- Dried Fruit & Nuts (Plums, Pears, Apples, Almonds)
- Candy & Baked Goods (Chocolate, Biscuits, Scones)
- Soups, Sauces, Condiments, Pasta, Dairy Products, Canned Meat, Oil, Grains, Fruit & Veg.
- Salespersons: Multiple, e.g., Mariya Sergienko (West), Andrew Cencini (East), Nancy Freehafer (North), Anne Larsen (South).
- Shipping Companies: A, B, C (frequent across regions).

Observed Patterns

- Highfrequency products: Coffee, Green Tea, Chocolate Biscuits Mix, Curry Sauce, Clam Chowder, Crab Meat.
- Unit Prices: Range from lowcost items (e.g., Green Tea at 2.99) to premium items (Marmalade at 81).
- Quantities: Orders vary widely (from ~10 units to 100+ units).

- Duplicate Entries: Some orders (e.g., Order ID 1003 for Dried Pears) appear multiple times, suggesting repeated or split shipments.

1. Task -:

- The "Shipping Bottleneck" Problem
Client Question: "Are we getting slower at shipping? Which shipping company is the most reliable?"
- Your Task: Calculate the Lead Time (Days between Order and Shipping). Identify which shipping company has the highest average delay.

Solution -:

Goal: Determine if shipping is slowing down and identify the most reliable partner.

- Steps:
 1. Converted Order_Date and Shipped_Date to date-time objects.
 2. Calculated Lead Time = Shipped_Date - Order_Date.
 3. Grouped by Shipper_Name to find the average delay.
- Insight: All three shipping companies have a consistent average lead time of 2.0 days. There is currently no delay bottleneck from the carriers' side.
- File: Task1_Shipping_Performance.csv

	A	B
1	Shipper Name	Average Lead Time
2	Shipping Company A	2
3	Shipping Company E	2
4	Shipping Company C	2

2. Task -:

- The "Salesperson Performance" Mystery
- Client Question: "We have sales reps in different regions. Is the 'West' region doing better because of the market, or because of the salesperson Mariya?"
- Your Task: Compare Salesperson revenue vs. their Region. Find the "Top Performer" and the "Underperformer."

Solution -:

Goal: Compare salesperson revenue against regions to find top performers and underperformers.

- Steps:
 1. Aggregated Sales by both Salesperson and Region.
 2. Ranked employees by total revenue.
- Insight: Anne Larsen (South) is the top performer with over \$72,000 in sales. In the West region, Mariya Sergienko is the primary driver, significantly outperforming her regional colleague, Jan Kotas.
- File: Task2_Sales_Performance.csv
- Visualization: Use a Clustered Bar Chart to show individual vs. regional performance.

	A	B	C
1	Salesperson	Region	Sales
2	Anne Larsen	South	72163.25
3	Andrew Cencini	East	54957.4
4	Nancy Freehafer	North	51325.75094
5	Laura Giussani	East	39943.51
6	Mariya Sergienko	West	39546.75094
7	Michael Neipper	North	38540
8	Robert Zare	West	32530.6
9	Jan Kotas	West	17041.5

3. Task -:

- The "Product Portfolio" Audit
- Client Question: "We sell everything from Beer to Dried Fruit. What is our 'Hero Category' that brings in the most money, and what should we stop selling?"
- Your Task: Rank Categories by total Sales. Look for highvolume but lowprice items that might be eating up warehouse space.

Solution-:

Goal: Identify "Hero" categories and low-performing items.

- **Steps:**
 1. Calculated total Sales, Quantity, and Average Price for each Category.
 2. Identified low-margin, high-volume items.
- **Insight: Beverages** and **Jams/Preserves** are your "Hero Categories." **Grains** is the lowest revenue category and should be reviewed for potential discontinuation.
- **File:** Task3_Product_Audit.csv
- Visualization: Create a Treemap where the size is "Revenue" and the color is "Quantity."

	A	B	C	D
1	Category	Sales	Quantity	Unit_Price
2	Beverages	58258.91094	2550	23.46533333
3	Jams, Preserves	51541	917	51.52631579
4	Sauces	50520	1263	40
5	Dairy Products	33129.6	952	34.8
6	Baked Goods & Mixe	22955.40094	2016	11.40606061
7	Condiments	20278	1207	16.6
8	Canned Meat	19117.6	1039	18.4
9	Dried Fruit & Nuts	19023.5	1075	21.71428571
10	Candy	18528.25	1385	13.09259259
11	Soups	16829.6	1744	9.65
12	Oil	13322.4	624	21.35
13	Pasta	11056.5	567	19.5
14	Fruit & Veg	6942	178	39
15	Grains	2884	412	7
16	Shipping	1662	66	23.33333333

4. Task-:

- The "Payment Preference" Analysis

- Client Question: "Do customers who pay by 'Credit Card' spend more than those who pay by 'Check'?"
- Your Task: Calculate the Average Order Value (AOV) for each payment type.

Solution:-

Goal: Determine if Credit Card users spend more on average (AOV).

- Steps:
 1. Grouped data by Payment_Type.
 2. Calculated the Average of the Sales column per group.
- Insight: Credit Card transactions have the highest AOV (\$1,285), followed by Checks. Cash has the lowest average basket size.
- File: Task4_Payment_Analysis.csv

	A	B
1	Payment Type	Average Order Value
2	Credit Card	1285.36327
3	Check	1192.1105
4	Cash	1052.169828

Report bi -:



